(a) The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone
(b) The questions can be answered by using either statement alone
(c) The question can be answered by using both the statements together, but cannot be answered using either statement alone
(d) The equation cannot be answered even by using both the statements together.

Ans. (a) : It is given that the age of P and Q is less than 100 and more than 10. so P and Q are two-digit numbers
Let age of P = ab and Q = ba.
From statement-I, P > Q
ab > ba
There are various possibilities
81 > 18
72 > 27
63 > 36
: :
This is not sufficient to tell the difference of age.
Hence, the question cannot be answered only using statement1.
From statement 2.
Let ab, is the larger number,

$$10a + b + 10b + a = \frac{11}{6}(10a + b - 10b - a)$$

$$11a + 11b = \frac{11}{6}(9a - 9b)$$

$$\Rightarrow \quad a + b = \frac{1}{6}(9a - 9b)$$

$$\Rightarrow \quad a + b = \frac{3}{2}(a - b)$$

Thus a = 5b
The only possible values are a = 5, b = 1
So, ab = 51
ba = 15
Difference of their age can be easily determined using this statement alone.
Hence, option (a) is correct.

44. A man statement is followed by four statements labelled P, Q, R and S.
Choose the ordered pair of the statements where the first statement implies the second, and the two statements are logically consistent with the main statement.
Main Statement:
Pradeep becomes either a Director or a Producer.
Statement P : Pradeep is a Director.
Statement Q : Pradeep is a Producer.
Statement R : Pradeep is not a Director.
Statement S : Pradeep is not a Producer.
Select the correct answer.
(a) SP only (b) RQ only
(c) Both SP and RQ (d) Neither SP nor RQ

Ans. (c) : Main statement Pradeep becomes either a Director or a producer.
This implies that pradeep can be one or the other, but not both.
(1) SP :
Statements : Pradeep is not a producer
Statements Pradeep is a director
If Pradeep is not Producer, he can still be a director, which is logically consistent with the main statement.
(2) PQ : Statement R : Pradeep is not a director
Statement Q : Pradeep is a producer.
If Pradeep is not a director, he can still be a producers, which is also logically consistent with the main statement.
Hence, SP and RQ both are fulfill the given Condition.

45. If a + b means a – b; a – b means a × b; a × b means a ÷ b; a ÷ b means a + b then what is the value of 10 + 30 – 100 × 50 ÷ 25? (Operations are to be replaced simultaneously)
(a) 15 (b) 0 (c) –15 (d) –25

Ans. (d) : Putting the given condition,
10 – 30 × 100 ÷ 50 + 25
= 10 – 30 × 2 + 25
= 10 – 60 + 25 = 35 – 60 = –25
Hence, options is the correct answer.

46. If P means 'greater than (>)'; Q means 'less than (≯)'; R means 'not greater than (<)'; S means 'not less than (≮)' and T means 'equal to (=)', then consider the following statements:
1. If 2x (S)3y and 3x(T)4z, then 9y(P)8z./यदि 2x (S)3y और 2x (S)3y तो 9y(P)8z
2. If x(Q)2y and y(R)z, then x(R)z.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : Statement -1 is incorrect–
2 × (S) 3y and 3x(T) 4z, then 9y(P) 8z
On placing symbols according to instructions,
2x ≥ 3y and 3x = 4z, then 9y > 8z

$$\Rightarrow \quad x \geq \frac{3}{2}y \quad \text{_____ (i)}$$

& 3x = 4z

$$x = \frac{4}{3}z \quad \text{_____ (ii)}$$

From eqn (i) & (ii)

$$\frac{4}{3}z \geq \frac{3}{2}y$$

$$8z \geq 9y$$

Hence, statement 1 is incorrect
• **Statement 2 is incorrect–**
x (Q) 2y and y(R)z, then x(R)Z
on placing symbols according to instruction,
x < 2y and y ≤ z, then x ≤ z
⇒ x < 2y _____ (i)
& y ≤ z or 2y ≤ 2z _____ (ii)
So, x < 2y ≤ 2z
x ≤2z,
Hence, statement 2 is incorrect.

47. If in a certain code, 'ABCD' is written as 24 and 'EFGH' is written as 1680, then how is 'IJKL' written in that code?
(a) 11880 (b) 11240
(c) 7920 (d) 5940

Ans. (a) : ∵ ABCD = 24
1 × 2 × 3 × 4
& EFGH = 1680
5 × 6 × 7 × 8 = 1680
So, IJKL = 9 × 10 × 11 × 12 = 11880

48. If in a certain code, 'POT' is written as ATOP and 'TRAP' is written as APART, then how is 'ARENA' written in that code?
(a) AARENA (b) AANREA
(c) AANEAR (d) AANERA

Ans. (d) : Reverse order of 'POP' = TOP
& code of POT will be ATOP
Reverse order of 'TRAP' = PART
& code of TRAP will be APART
Same as,
Reverse order of ARENA = ANERA
& code of ARENA will be AANERA

49. What will come in place of * in the sequence 3, 15, 39, 84 * 258?
(a) 150 (b) 155 (c) 160 (d) 176

Ans. (b) : The given sequence is 3, 14 , 39, 84, x, 258
The difference between is 11, 25, 45, x – 84, 258 – x
The difference of this difference is 14, 20, x – 129, 342 – 2x
The difference of difference follows a gap of 6 after starting from 14 pattern for difference of difference is 14, 20, 26, 32
Hence, x – 1296 = 26
x = 155

50. In some code, letters P, Q, R, S, T represent numbers 4, 5, 10, 12, 15. It is not known which letter represents whic number. If Q – S = 2S and T = R + S + 3, then what is the value of P + R – T?
(a) 1 (b) 2
(c) 3
(d) Cannot be determined due to insufficient data

Ans. (d) : As Q – 5 = 25
Q = 35
Possible values of Q = 12, 15
Now as T = R + S + 3,
So, T = 15,
R, S belong to (4, 5)
AS T = 15, So Q = 12 which implies S = 4, which implies R = 5
So, P = 10 (only remaining number from the list)
So, P + R – T = 10 + 5 – 15 = 0

Directions for the following 3 (three) items:
Read the following three passages and answer the items that follow the passages. Your answers to these items should be based on the passage only.

Passage–1

In a robust democracy, reality, howsover inconvenient it may be, finds its expression both in the speech of political leaders and the other social forms of assertion. The existence of the real has to be transparent, both through its circulation in and by the media as well as its argumentative articulation in deliberative democracy. A normatively responsible media through its communication effect has the responsibility to circulate the content of reality without distortion.

51. Which one of the following statements best reflects the crux of the above passage?
(a) Responsible media should not distort the real in an ideal democracy.
(b) Fake news seems inherent in the life of an ideal democracy.
(c) There should not be any kind of restrictions on the freedom of expression in an ideal democracy.
(d) Irresponsible media and political leaders cannot be effectively controlled in an ideal democracy.

Ans. (a) : Options (a) best reflects the crux of the above passage. The passage forces the importance of a normatively responsible media in the sobust democracy, where reality must be transparently circulated and articulated without distortion.
• Options (b) is incorrect, This statement contradicts the passage, which emphasizes the need for responsible media to avoid distortion implying the take new is not desirable in a ideal democracy.
• Options (c) is not correct, while the passage does discuses freedom of expression, it focus is more on the accuracy and responsibility of the media in conveying reality, rather than on the absence of restrictions.
• Options (d) is incorrect, because the passage emphasizes the need for media responsibility in circulating reality, rather than addressing the control of irresponsibility within media and political leadership.

Passage–2

Now-a-days there is a growing trend to use interconnected home devices. As consumers increasingly network their homes, the connected home device manufacturers and service providers will seek to overcome ''thin profit margins by gathering more of our personal data–with or without our agreement–turning the home into a corporate storefront''. Corporate markters will have powerful incentives to observe consumer behaviour to understand the buying needs and preferences of the device owners.

52. Which one of the following statements best reflects the most logical, rational and practical message implied by the passage?
(a) Knowledge of consumer behaviour leads to more capital expenditure in manufacturing sector.
(b) Knowledge of consumer behaviour stimulates the growth of commerce and trade and thus helps in the overall economic development of the country.
(c) Interconnected devices give a lot of comfort to home users and improve the overall quality of life.
(d) Interconnected devices can be at security risk and home users may have privacy risk.

Ans. (d) : Options (a) is incorrect because the passage mainly focuses on the implication of gathering personal data and marketing strategies, rather than discussing increased capital expenditure in manufacturing sector.
• Options (b) while it is true that knowledge of consumer behaviour stimulates the growth of commerce, the passage does not connect this to economic development.
• Options (c) is incorrect, because although interconnected devices may enhance comfort and quality of life for users, the passage does not emphasize this aspect as a central theme.
• Options (d) is the correct answer because the passage mentions the potential risks associated with interconnect devices, such as security risks and privacy concerns for home users. It discusses how manufactures and service providers may collect personal data without full consumer agreement which can lead to these risks.

Passage–3

Green growth involves rethinking growth strategies with regrad to the impacts on environmental sustainability and the environmental resources available to poor and vulnerable groups. In rethinking growth, we need to focus on the current reality of a resource-constrained world. Resource intensive and, in particular energy intensive processes will need to make way for more efficient and resource frugal development strategies if we are to avoid an economic dead end or a world in which only a small elite is able to enjoy affluence in the midst of a sea of poverty.